

### 1. Textbook problems

- 3.3 A tornado rotates with constant angular velocity  $\omega$ . Show that the surface pressure at the center of the tornado is given by

$$p = p_0 \exp\left(\frac{-\omega^2 r_0^2}{2RT}\right)$$

where  $p_0$  is the surface pressure at a distance  $r_0$  from the center and  $T$  is the temperature (assumed constant). If the temperature is 288 K and pressure and wind speed at 100 m from the center are 1000 hPa and  $100 \text{ m s}^{-1}$ , respectively, what is the central pressure?

Ans:

Letting  $r$  be the distance from center (the axis of rotation) of the tornado, and shifting to height coordinates. The cyclostrophic balance flow can be expressed as

$$-\frac{\partial \Phi}{\partial n} - \frac{v^2}{r} = 0 \implies \frac{v^2}{r} = -\frac{\partial \Phi}{\partial n} = \frac{1}{\rho} \frac{\partial p}{\partial r}$$

where the  $n$  direction is the normal direction of the airflow, which, for this problem (tornado), is in the  $r$  direction, and  $d\Phi = dp/\rho$ .

But  $v = \omega r$  and  $1/\rho = RT/p$ , so that

$$\frac{\partial \ln p}{\partial r} = \frac{\omega^2 r}{RT}$$

Integrating with respect to  $r$ :

$$\begin{aligned} \int_{p_0}^p d \ln p &= \left(\frac{\omega^2}{RT}\right) \int_{r_0}^0 r dr \\ \ln\left(\frac{p}{p_0}\right) &= \left(\frac{\omega^2}{RT}\right) \left(\frac{-r^2}{2}\right) \Big|_{r_0}^0 \\ p &= p_0 \exp\left(\frac{-\omega^2 r_0^2}{2RT}\right) \end{aligned}$$

Substitute  $T = 288K$ ,  $v = r\omega = 100 \text{ m s}^{-1}$ ,  $p_0 = 1000 \text{ hPa}$  into equation, we have

$$p \approx 941.3 \text{ hPa}$$

- 3.5 Determine the maximum possible ratio of the normal anticyclonic gradient wind speed to the geostrophic wind speed for the same pressure gradient.

Ans:

From Gradient wind equation

$$V = \frac{-fR}{2} \pm \sqrt{\frac{f^2 R^2}{4} - R \frac{\partial \Phi}{\partial n}}$$

A regularly high takes the negative root in Gradient wind equation and has  $R < 0$ , we have

$$V_{grad} = \frac{-fR}{2} \pm \sqrt{\frac{f^2 R^2}{4} - fRV_g}$$

Thus,

$$(V_{grad})_{max} = \frac{-fR}{2}$$

That implies  $\frac{f^2 R^2}{4} - fRV_g = 0$  for the maximum gradient wind speed.

Thus, for the maximum gradient wind,

$$V_g = \frac{fR}{4} = \frac{1}{2}(V_{grad})_{max}$$

so that maximum gradient wind is twice the geostrophic wind speed in a normal anticyclone.

- 3.6 Show that the geostrophic balance in isothermal coordinates may be written:

$$fV_g = k \times \nabla_T(RT \ln p + \Phi)$$

Ans:

From the figure,

$$\frac{p_B - p_a}{\delta x} = \frac{p_C - p_A}{\delta x} + \frac{p_B - p_C}{\delta Z} \left( \frac{\delta Z}{\delta x} \right)_T$$

Thus,

$$\left( \frac{\delta p}{\delta x} \right)_T = \left( \frac{\delta p}{\delta x} \right)_Z + \left( \frac{\delta p}{\delta Z} \right) \left( \frac{\delta Z}{\delta x} \right)_T$$

From geostrophic wind equation

$$f v_g = \frac{1}{\rho} \left( \frac{\partial p}{\partial x} \right)_Z$$

And we use  $\frac{\partial p}{\partial Z} = -\rho g_0$ , then

$$f v_g = \frac{1}{\rho} \left( \frac{\partial p}{\partial x} \right)_T + g_0 \left( \frac{\partial Z}{\partial x} \right)_T$$

If we eliminate  $\rho$  using  $1/\rho = RT/p$ , and use that  $\Phi = g_0 Z$ , we obtain

$$\begin{aligned} f v_g &= \frac{RT}{p} \left( \frac{\partial p}{\partial x} \right)_T + \left( \frac{\partial \Phi}{\partial x} \right)_T \\ &= \frac{\partial}{\partial x} (RT \ln p + \Phi)_T \end{aligned}$$

Similar derivation to  $u_g$

- 3.12 Suppose that a vertical column of the atmosphere at 43°N is initially isothermal from 900 to 500 hPa. The geostrophic wind is 10  $ms^{-1}$  from the south at 900 hPa, 10  $m s^{-1}$  from the west at 700 hPa, and 20  $ms^{-1}$  from the west at 500 hPa. Calculate the mean horizontal temperature gradients in the two layers 900 to 700 hPa and 700 to 500 hPa. Compute the rate of advective temperature change in each layer. How long would this advection pattern have to persist in order to establish a dry adiabatic lapse rate between 600 and 800 hPa? (Assume that the lapse rate is constant between 900 and 500 hPa and that the 800- to 600-hPa layer thickness is 2.25 km.)

From thermal wind equation:

$$\vec{v}_T = -\frac{R}{f} (\hat{k} \times \nabla \bar{T}) \ln(P_0/P_1)$$

where  $\vec{v}_T \equiv \vec{v}_g(P_1) - \vec{v}_g(P_0)$

Expand the equation, we have

$$u_T = -\frac{R}{f} \left( \frac{\partial \bar{T}}{\partial y} \right) \ln \left( \frac{P_0}{P_1} \right) \quad v_T = \frac{R}{f} \left( \frac{\partial \bar{T}}{\partial x} \right) \ln \left( \frac{P_0}{P_1} \right)$$

Rearrange the equation

$$\frac{\partial \bar{T}}{\partial y} = \frac{-f u_T}{R \ln(P_0/P_1)} \quad \frac{\partial \bar{T}}{\partial x} = \frac{f v_T}{R \ln(P_0/P_1)}$$

For layer 900-700 hPa:  $u_T = 10m/s$ ,  $v_T = -10m/s$

$$\frac{\partial T}{\partial x} = \frac{\partial \bar{T}}{\partial x} = -1.38 \times 10^{-5} \text{ } ^\circ C/m = -1.38^\circ C/100km$$

For layer 700-500 hPa:  $u_T = 10m/s$ ,  $v_T = 0m/s$

$$\frac{\partial T}{\partial x} = 0 \quad \frac{\partial T}{\partial y} = -1.03^\circ C/100km$$

The advective rate of temperature change is

$$\frac{\partial T}{\partial t} = \cancel{\frac{dT}{dt}} - u \frac{\partial T}{\partial x} - v \frac{\partial T}{\partial y}$$

since there is no local heating term.

The average wind in the 900-700hPa layer is  $\bar{u} = \bar{v} = 5m/s$

and in the 700-500hPa layer it is  $\bar{u} = 15m/s, \bar{v} = 0$  Thus, for the 900-700 hPa layer

$$\frac{\partial \bar{T}}{\partial t} = 1.39 \times 10^{-4} \text{ } ^\circ C/s \approx 0.5^\circ C/hr$$

for the 700-500 hPa layer,

$$\frac{\partial \bar{T}}{\partial t} = 0$$

Therefore, the temperature difference between 800 hPa and 600 hPa increases by  $0.5^\circ\text{C}/\text{hr}$ .

According the topic, the thickness of layer is 2.25km and that the adiabatic lapse rate is  $9.8^\circ\text{C}/\text{km}$

When establish a dry adiabatic lapse rate, the temperature difference between 800 hPa and 600 hPa is  $9.8^\circ\text{C}/\text{km} \times 2.25 \text{ km}$ .

Thus, the time the layer establish a dry adiabatic lapse rate is

$$\frac{9.8^\circ\text{C}/\text{km} \times 2.25 \text{ km}}{0.5^\circ\text{C}/\text{hr}} = 44.1 \text{ hr}$$

- 3.20 The following wind data were received from 50 km to the east, north, west, and south of a station, respectively:  $90^\circ$ ,  $10 \text{ ms}^{-1}$ ;  $120^\circ$ ,  $4 \text{ ms}^{-1}$ ;  $90^\circ$ ,  $8 \text{ ms}^{-1}$ ; and  $60^\circ$ ,  $4 \text{ ms}^{-1}$ . Calculate the approximate horizontal divergence at the station.

Ans:

Let  $\delta x = \delta y = 50 \text{ km}$

Reorganize the data in the table below

| station  | speed(m/s) | direction   | u(m/s)       | v(m/s) |
|----------|------------|-------------|--------------|--------|
| east(A)  | 10         | $90^\circ$  | -10          | 0      |
| north(B) | 4          | $120^\circ$ | $-2\sqrt{3}$ | 2      |
| west(C)  | 8          | $90^\circ$  | -8           | 0      |
| south(D) | 4          | $60^\circ$  | $-2\sqrt{3}$ | -2     |

The horizontal divergence is

$$D = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$$

we obtain the finite difference formula:

$$D \approx \frac{u_A - u_C}{2\delta x} + \frac{v_B - v_D}{2\delta y} = \frac{(-10 + 8)}{10^5} + \frac{(2 + 2)}{10^5} = 2 \times 10^{-5} \text{ s}^{-1}$$

- 3.21 Suppose that the wind speeds given in Problem 3.20 are each in error by  $\pm 10\%$ . What would be the percentage error in the calculated horizontal divergence in the worst case?

Ans:

For the worst case, let  $u_A = -11 \text{ ms}^{-1}$ ,  $u_C = -7.2 \text{ ms}^{-1}$ ,  $v_B = 1.8 \text{ ms}^{-1}$ ,  $v_D = -1.8 \text{ ms}^{-1}$ , then

$$D = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{(-11 + 7.2)}{10^5} + \frac{(1.8 + 1.8)}{10^5} = -2 \times 10^{-6}$$

The error is

$$\frac{2 \times 10^{-6} - 2 \times 10^{-5}}{2 \times 10^{-5}} = -1.1 \Rightarrow 110\%$$

- 3.22 The divergence of the horizontal wind at various pressure levels above a given station is shown in the following table.

| Pressure (hPa) | $\nabla \cdot \mathbf{V} (\times 10^{-5} \text{ s}^{-1})$ |
|----------------|-----------------------------------------------------------|
| 1000           | +0.9                                                      |
| 850            | +0.6                                                      |
| 700            | +0.3                                                      |
| 500            | 0.0                                                       |
| 300            | -0.6                                                      |
| 100            | -1.0                                                      |

Compute the vertical velocity at each level assuming an isothermal atmosphere with temperature 260 K and letting  $w = 0$  at 1000 hPa.

Ans:

From the continuous function

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial p} = 0$$

Integrate with respect to pressure from bottom to top, we have

$$\int_{P_0}^P \left[ \frac{\partial w}{\partial p} = - \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \right] dP$$

Simplify the equation, we get vertical motion equation

$$\omega(P) = \omega(P_0) + \left( \frac{\partial \bar{u}}{\partial x} + \frac{\partial \bar{v}}{\partial y} \right) (P_0 - P) = \omega(P_0) + (\nabla \cdot \bar{V})(P_0 - P)$$

where in this case the vertical average of the divergence in each layer must be estimated from averaging the top and bottom values

$$(\nabla \cdot \bar{V}) = \frac{1}{2}[(\nabla \cdot V)_{p_0} + (\nabla \cdot V)_{p_1}]$$

The averaged horizontal wind (in  $\times 10^{-5} s^{-1}$ ) are

$$\nabla \cdot \bar{V}(1000 - 850 hPa) = 0.75$$

$$\nabla \cdot \bar{V}(850 - 700 hPa) = 0.45$$

$$\nabla \cdot \bar{V}(700 - 500 hPa) = 0.15$$

$$\nabla \cdot \bar{V}(500 - 300 hPa) = -0.3$$

$$\nabla \cdot \bar{V}(300 - 100 hPa) = -0.8$$

If  $\omega(1000 hPa) = 0$ , then integrating upward gives

$$\omega(850 hPa) = 11.25 \times 10^{-2} Pa s^{-1}$$

$$\omega(700 hPa) = 18.00 \times 10^{-2} Pa s^{-1}$$

$$\omega(500 hPa) = 21.00 \times 10^{-2} Pa s^{-1}$$

$$\omega(300 hPa) = 15.00 \times 10^{-2} Pa s^{-1}$$

$$\omega(100 hPa) = -1.0 \times 10^{-2} Pa s^{-1}$$

Now  $w \simeq \frac{\omega}{-\rho g} = -\frac{RT}{pg}\omega$ , where  $T = 260 K$  and  $g \sim 9.8 m/s^2$

Thus, from the above expressions we estimate the vertical velocity at each levels as

$$w(850 hPa) = -1.0 cm s^{-1}$$

$$w(700 hPa) = -2.0 cm s^{-1}$$

$$w(500 hPa) = -3.3 cm s^{-1}$$

$$w(300 hPa) = -3.8 cm s^{-1}$$

$$w(100 hPa) = +0.8 cm s^{-1}$$

- 4.2 What is the circulation about a square of 1000 km on a side for an easterly (i.e., westward flowing) wind that decreases in magnitude toward the north at a rate of  $10 ms^{-1}$  per 500 km? What is the mean relative vorticity in the square?

Ans:

By definition of relative vorticity

$$\zeta \equiv \hat{k} \cdot \vec{\omega} = \hat{k} \cdot (\nabla \times \vec{V}) = \frac{\partial u}{\partial y} - \frac{\partial v}{\partial x}$$

Substitute the condition of topic, we have

$$\zeta = \frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} = 0 \left( \frac{10 ms^{-1}}{5 \times 10^5 m} \right) = -2 \times 10^{-5} s^{-1}$$

And the circulation is

$$C \equiv \oint_{\Gamma} \vec{V} \cdot d\vec{\ell} = \iint (\nabla \times V) \cdot \hat{n} dA = \iint \zeta dA \sim \zeta_{mean}(\Delta A)$$

$$C = (-2 \times 10^{-5} s^{-1}) (10^6 m)^2 = -2 \times 10^7 m^2 s^{-1}$$

- 4.4 An air column at  $60^\circ N$  with  $\zeta = 0$  initially stretches from the surface to a fixed tropopause at 10 km height. If the air column moves until it is over a mountain barrier 2.5 km high at  $45^\circ N$ , what is its absolute vorticity and relative vorticity as it passes the mountain top, assuming that the flow satisfies the barotropic potential vorticity equation?

Ans:

By conservation of potential vorticity

$$PV = \frac{\zeta_\theta + f}{\rho \delta z} = \frac{\zeta_\theta + f}{H} = Constant$$

According to topic,  $\zeta = 0$ , initially, so

$$\begin{aligned}\frac{f_{initial}}{H_{initial}} &= \frac{(\zeta + f)_{final}}{H_{final}} \\ \frac{2\Omega \sin(60^\circ)}{10 \text{ km}} &= \frac{(\zeta + 2\Omega \sin(45^\circ))}{10 - 2.5 \text{ km}} \\ \Rightarrow \zeta_{final} &= -8.4 \times 10^{-6} \text{ s}^{-1}\end{aligned}$$

- 4.5 Find the average vorticity within a cylindrical annulus of inner radius 200 km and outer radius 400 km if the tangential velocity distribution is given by  $V = A/r$ , where  $A = 10^6 \text{ m}^2 \text{ s}^{-1}$  and  $r$  is in meters. What is the average vorticity within the inner circle of radius 200 km?

Ans:

From the definition of circulation

$$C \equiv \oint_{\Gamma} \mathbf{V} \cdot d\mathbf{\ell} = \int_0^{2\pi} \mathbf{V} r d\theta = 2\pi r \mathbf{V}$$

For the outer circle

$$C_o = (2\pi r_o)(10^6/r_o) = 2\pi \times 10^6 \text{ m}^2 \text{ s}^{-1}$$

,and for the inner circle

$$C_i = (2\pi r_i)(-10^6/r_i) = -2\pi \times 10^6 \text{ m}^2 \text{ s}^{-1}$$

Note that  $C_i$  is taken by going clockwise around the circle, or opposite to direction of  $V$ .

Thus, since  $C_o + C_i = 0$ , the vorticity vanishes in the annular region.

The mean vorticity in the inner cylinder is given by the circulation divided by area for the inner circle:

$$\zeta_m = \frac{C}{A} = \frac{2\pi \times 10^6 \text{ m}^2 \text{ s}^{-1}}{\pi(2 \times 10^5)^2 \text{ m}^2} = 5 \times 10^{-5} \text{ s}^{-1}$$

Another solution:

Under natural coordinate

$$\zeta = \frac{V}{R_s} - \frac{\partial V}{\partial n} = \frac{V}{R} + \frac{\partial V}{\partial r} = \frac{A}{R^2} + \left(-\frac{A}{R^2}\right) = 0$$

Thus, the average vorticity within a cylindrical annulus (200km to 400km) is

$$\zeta = 0 \text{ s}^{-1}$$

- 4.7 Compute the rate of change of circulation about a square in the (x, y) plane with corners at (0, 0), (0, L), (L, L), and (L, 0) if temperature increases eastward at a rate of  $1^\circ\text{C}$  per 200 km and pressure increases northward at a rate of 1 hPa per 200 km. Let  $L = 1000$  km and the pressure at the point (0, 0) be 1000 hPa.

Ans:

From the circulation theorem

$$\frac{dC}{dt} = - \oint \frac{-dp}{\rho} = \oint \frac{-RT}{p} dp$$

But  $\ln p$  changes only for the North-South segments of the square, so if  $\delta T$  and  $\delta p$  are the temperature and pressure differences,

$$\frac{dC}{dt} = -R\delta T \ln \left[ \frac{p_0 + \delta p_0}{p_0} \right] = 287 \times 5 \times \ln \left( \frac{1005}{1000} \right) = -7.15 \text{ m}^2 \text{ s}^{-2}$$

- 4.8 Verify the identity (4.14) by expanding the vectors in Cartesian components.

$$(\mathbf{V} \cdot \nabla) \mathbf{V} = \nabla \left( \frac{\mathbf{V} \cdot \mathbf{V}}{2} \right) + \zeta \mathbf{k} \times \mathbf{V}$$

Ans:

In this question, we only care about 2-Dimension vector

$$\begin{aligned}(\mathbf{V} \cdot \nabla) \mathbf{V} &= \left[ (u, v) \cdot \left( \frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right) \right] \mathbf{V} = \left( u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} \right) (u, v) \\ &= \mathbf{i} \left[ u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right] + \mathbf{j} \left[ u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right]\end{aligned}$$

$$\begin{aligned}\nabla \left( \frac{\mathbf{V} \cdot \mathbf{V}}{2} \right) &= \nabla \left( \frac{u^2 + v^2}{2} \right) = \mathbf{i} \frac{\partial}{\partial x} \left( \frac{u^2 + v^2}{2} \right) + \mathbf{j} \frac{\partial}{\partial y} \left( \frac{u^2 + v^2}{2} \right) \\ &= \mathbf{i} \left[ u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial x} \right] + \mathbf{j} \left[ u \frac{\partial u}{\partial y} + v \frac{\partial v}{\partial y} \right] \\ \mathbf{k} \times \mathbf{V} \zeta &= \mathbf{k} \times (\mathbf{i}u + \mathbf{j}v) \left[ \frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right]\end{aligned}$$

But  $\mathbf{k} \times \mathbf{i} = \mathbf{j}$  and  $\mathbf{k} \times \mathbf{j} = -\mathbf{i}$ . Thus,

$$\mathbf{k} \times \mathbf{V} \zeta = \mathbf{j} \left[ u \frac{\partial u}{\partial x} - v \frac{\partial v}{\partial y} \right] - \mathbf{i} \left[ v \frac{\partial u}{\partial x} - u \frac{\partial v}{\partial y} \right]$$

Combine the equation above, we have

$$(\mathbf{V} \cdot \nabla) \mathbf{V} = \nabla \left( \frac{\mathbf{V} \cdot \mathbf{V}}{2} \right) + \zeta \mathbf{k} \times \mathbf{V}$$

, which was to be proved.

- 4.11 A cyclonic vortex is in cyclostrophic balance with a tangential velocity profile given by the expression  $V = V_0(r/r_0)^n$  where  $V_0$  is the tangential velocity component at the distance  $r_0$  from the vortex center. Compute the circulation about a streamline at radius  $r$ , the vorticity at radius  $r$ , and the pressure at radius  $r$ . (Let  $p_0$  be the pressure at  $r_0$  and assume that density is a constant.)

Ans:

$$C = 2\pi V r = 2\pi V_0 \frac{r^{n+1}}{r_0^n}$$

the vorticity in polar coordinate is

$$\zeta = \frac{1}{r} \frac{\partial(rV)}{\partial r} = \frac{V_0}{r_0^n r} \frac{\partial(r^{n+1})}{\partial r} = (n+1) \frac{V_0}{r_0^n} r^{n-1} = (n+1) \frac{V}{r}$$

The cyclostrophic flow implies

$$-\frac{\partial \Phi}{\partial n} - \frac{V^2}{R} = 0 \implies \frac{V^2}{r} = \frac{1}{\rho} \frac{\partial p}{\partial r} = \frac{V_0^2 r^{2n-1}}{r_0^{2n}}$$

Move  $dr$  to other side and integrate both sides

$$\int_{p_0}^p \frac{dp}{\rho} = \int \frac{V_0^2 r^{2n-1}}{r_0^{2n}} dr$$

so that,

$$\frac{p - p_0}{\rho} = \frac{V_0^2}{r_0^{2n}} \left( \frac{r^{2n} - r_0^{2n}}{2n} \right) = \frac{V^2 - V_0^2}{2n}$$

- 4.12 A westerly zonal flow at  $45^\circ$  is forced to rise adiabatically over a north-south-oriented mountain barrier. Before striking the mountain, the westerly wind increases linearly toward the south at a rate of 10 m s<sup>-1</sup> per 1000 km. The crest of the mountain range is at 800 hPa and the tropopause, located at 300 hPa, remains undisturbed. What is the initial relative vorticity of the air? What is its relative vorticity when it reaches the crest if it is deflected  $5^\circ$  latitude toward the south during the forced ascent? If the current assumes a uniform speed of 20 m s<sup>-1</sup> during its ascent to the crest, what is the radius of curvature of the streamlines at the crest?

Ans:

Assume the pressure of ground is 1000hPa By the potential vorticity conservation

$$\frac{f_0 + \zeta_0}{\delta p_0} = \frac{f_1 + \zeta_1}{\delta p_1}$$

The initial condition is

$$\zeta_0 = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = \frac{10 \text{ m/s}}{10^6 \text{ m}} = 10^{-5} \text{ s}^{-1}$$

$$f_0 = 2\Omega \sin(45^\circ) = 1.03 \times 10^{-4} \text{ s}^{-1}$$

$$\delta p_0 = 700 \text{ hPa}$$

The final condition is

$$f_1 = 2\Omega \sin(40^\circ) = 9.37 \times 10^{-5} \text{ s}^{-1}$$

$$\delta p_1 = 500 \text{ hPa}$$

Substitute into potential vorticity conservation, we find

$$\zeta_1 = -1.29 \times 10^{-5} \text{ s}^{-1}$$

For a uniform flow the vorticity is entirely due to curvature so that

$$\zeta_1 = \frac{V}{R} \implies R = \frac{V}{\zeta_1} = -1.55 \times 10^6 \text{ m} (-1550 \text{ km})$$

- 4.14 (a) How far must a zonal ring of air initially at rest with respect to Earth's surface at  $60^\circ$  latitude and 100km height be displaced latitudinally in order to acquire an easterly (east to west) component of  $10 \text{ m s}^{-1}$  with respect to Earth's surface?

Ans:

we can consider conservation of angular momentum for small changes  $\delta R$  in the distance to the axis of rotation of Earth to make an approximate solution

$$\Omega R^2 = \left[ \Omega + \frac{\delta u}{(R + \delta R)} \right] (R + \delta R)^2$$

,where  $\delta u = 10 \text{ m s}^{-1}$ , zonal wind under angular momentum conservation.

Expanding gives

$$\lim_{\delta \rightarrow 0} \implies \delta u = -2\Omega \delta R - \frac{u}{R} \delta R \approx -2\Omega \delta R$$

For latitudinal displacement on  $60^\circ$  latitude, we find

$$\delta R = \delta y \sin \phi \implies \delta y = \frac{\delta u}{2\Omega \sin(60^\circ)} = -79 \text{ km}$$

- (b) To what height must it be displaced vertically in order to acquire the same velocity? Assume a frictionless atmosphere.

Ans:

While for vertical displacement on  $60^\circ$  latitude, we have

$$\delta R = \delta z \cos \phi \implies \delta z = \frac{-\delta u}{2\Omega \cos(60^\circ)} = 138 \text{ km}$$

- 4.15 The horizontal motion within a cylindrical annulus with permeable walls of inner radius 10 cm, outer radius 20 cm, and 10-cm depth is independent of height and azimuth and is represented by the expressions  $u = 7 - 0.2r$ ,  $v = 40 + 2r$ , where  $u$  and  $v$  are the radial and tangential velocity components in  $\text{cm s}^{-1}$ , positive outward and counterclockwise, respectively, and  $r$  is distance from the center of the annulus in centimeters. Assuming an incompressible fluid, find

- (a) Circulation about the annular ring

Ans:

$$C = 2\pi r_0 v(r_0) - 2\pi r_1 v(r_1) = 2\pi[20(40 + 2 \times 20) - 10(40 + 2 \times 10)] = 6283 \text{ cm}^2/\text{s}$$

- (b) Average vorticity within the annular ring

Ans:

$$\zeta = C/A = \frac{2\pi \times 10^3}{\pi r_0^2 - \pi r_1^2} = \frac{2\pi \times 10^3}{300\pi} = 6.67 \text{ s}^{-1}$$

- (c) Average divergence within the annular ring

Ans:

From the divergence theorem

$$\begin{aligned} \nabla \cdot \mathbf{V} &= \frac{1}{r} \frac{\partial}{\partial r}(ru) \\ \implies \iint \nabla \cdot \mathbf{V} dA &= \oint_{r_0} \mathbf{V} \cdot \mathbf{n} ds - \oint_{r_1} \mathbf{V} \cdot \mathbf{n} ds = A(\nabla \cdot \mathbf{V})_{mean} \end{aligned}$$

and  $A = \pi(r_0^2 - r_1^2)$ , so that

$$(\nabla \cdot \mathbf{V})_{mean} = \frac{2\pi r_0 u(r_0) - 2\pi r_1 u(r_1)}{\pi(r_0^2 - r_1^2)} = 0.0667 \text{ s}^{-1}$$

- (d) Average vertical velocity at the top of the annulus if it is zero at the base

Ans:

Now from the continuous equation

$$\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = -\frac{\partial w}{\partial z}$$

$$\frac{\partial w}{\partial z}_{mean} = -(\nabla \cdot \mathbf{V})_{mean}$$

so that integrating in height gives:

$$w_{mean} = -H(\nabla \cdot \mathbf{V})_{mean} = -0.667 \text{ cm s}^{-1}$$

at  $z = H$

## 2. Inclass problems.

1. To gain an appreciation for the possible errors involved in using the curvature of the streamlines instead of the curvature of the trajectories in the gradient wind equation, it is necessary to investigate the relationship between the curvature of the trajectories and streamlines for a moving pressure system. Now consider an idealized circular pattern of height contours moving at a constant velocity  $C$ .

- (a) Please follow equation (3.20) to (3.23) and derive (3.24):

$$\frac{D\beta}{Ds} = \frac{1}{R_t} \quad \frac{\partial\beta}{\partial s} = \frac{1}{R_s} \quad (3.20)$$

$$\frac{D\beta}{Dt} = \frac{D\beta}{Ds} \frac{Ds}{Dt} = \frac{V}{R_t} \quad (3.21)$$

$$\frac{D\beta}{Dt} = \frac{\partial\beta}{\partial t} + V \frac{\partial\beta}{\partial s} = \frac{\partial\beta}{\partial t} + \frac{V}{R_s} \quad (3.22)$$

$$\frac{\partial\beta}{\partial t} = V \left( \frac{1}{R_t} - \frac{1}{R_s} \right) \quad (3.23)$$

$$R_t = R_s \left( 1 - \frac{C \cos \gamma}{V} \right)^{-1} \quad (3.24)$$

Ans:

We let  $\beta(x, y, t)$  designate the angular direction of the wind at each point on an isobaric surface

And  $R_t$  and  $R_s$  designate the radius of curvature of the trajectories and streamlines, respectively.

Since  $\delta s = R\delta\beta$  so that in the limit  $\delta s \rightarrow 0$

$$\frac{D\beta}{Ds} = \frac{1}{R_t}$$

,where  $\frac{D\beta}{Ds}$  means the rate of change of wind direction along a trajectory (positive for counterclockwise turning)

$$\frac{\partial\beta}{\partial s} = \frac{1}{R_s}$$

,where  $\frac{\partial\beta}{\partial s}$  is the rate of change of wind direction along a streamline at any instant. Thus, the rate of change of wind direction following the motion by chain rule is

$$\frac{D\beta}{Dt} = \frac{D\beta}{Ds} \frac{Ds}{Dt} = \frac{V}{R_t}$$

or, after expanding the total derivative,

$$\frac{\partial\beta}{\partial t} = V \left( \frac{1}{R_t} - \frac{1}{R_s} \right)$$

Combining (3.21) and (3.22), we obtain a formula for the local turning of the wind:

$$\frac{\partial\beta}{\partial t} = V \left( \frac{1}{R_t} - \frac{1}{R_s} \right)$$

The relationship between  $R_t$  and  $R_s$  in midlatitude synoptic systems can be determined easily for an idealized circular pattern of height contours moving at a constant velocity  $\mathbf{C}$ .

In this case the local turning of the wind is entirely due to the motion of the streamline pattern so that.

$$\frac{\partial\beta}{\partial t} - \mathbf{C} \cdot \nabla\beta = -C \frac{\partial\beta}{\partial s} \cos \gamma - \frac{C}{R_s} \cos \gamma$$



where  $\gamma$  is the angle between the streamlines (height contours) and the direction of motion of the system. Substituting and rearranging the preceding equation

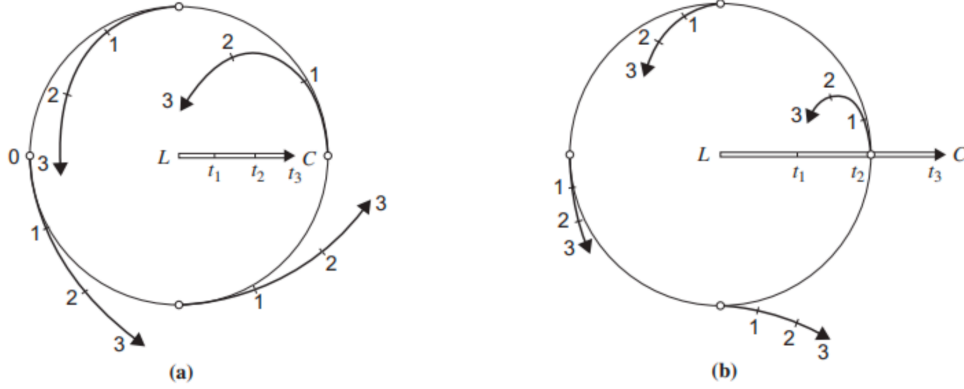
$$-C \frac{\partial \beta}{\partial s} \cos \gamma = \left( V \frac{1}{R_t} - \frac{1}{R_s} \right)$$

$$\frac{-C \cos \gamma}{R_s} = V \left( \frac{1}{R_t} - \frac{1}{R_s} \right)$$

we obtain the desired relationship between the curvature of the streamlines and the curvature of the trajectories:

$$R_t = R_s \left( 1 - \frac{C \cos \gamma}{V} \right)$$

- (b) Discuss the examples in Figure 3.7 and compare the different patterns between (a)  $V = 2C$  and (b)  $2V = C$ .



**FIGURE 3.7** Trajectories for moving circular cyclonic circulation systems in the Northern Hemisphere with (a)  $V = 2C$  and (b)  $2V = C$ . Numbers indicate positions at successive times.  $L$  designates a pressure minimum.

Ans:

This shows the curvature forms of the trajectories of air parcels initially located in different positions around the cyclonic system in two situations.

(a) is for the case of a wind speed greater than the speed of movement of the height contours and (b) is for the case of a wind speed less than the speed of movement of the height contours.

Assumed for simplicity that the wind speed does not depend on the distance from the center of the system.

On figure (b), there is a region south of the low center where the curvature of the trajectories is opposite that of the streamlines.

Since synoptic-scale pressure systems usually move at speeds comparable to the wind speed, the gradient wind speed computed on the basis of the curvature of the height contours is often no better an approximation to the actual wind speed than the geostrophic wind. In fact, the actual gradient wind speed will vary along a height contour with the variation of the trajectory curvature.

2. Show that the vorticity in natural coordinates is given by

$$\vec{\omega} = \nabla \times \vec{v} = \nabla \times (V\hat{t}) = \left( \frac{V}{R} - \frac{\partial V}{\partial n} \right) \hat{k}$$

Ans:

$$\nabla \times \vec{v} = \nabla \times (\vec{V}\hat{t}) = V(\nabla \times \hat{t}) + \nabla V \times \hat{t} = V(\nabla \times \hat{t}) - \hat{t} \times \nabla V$$

$$\nabla \times \hat{t} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \cos \psi & \sin \psi & 0 \end{vmatrix} = \frac{\partial}{\partial x} \sin \psi \hat{k} + \frac{\partial}{\partial z} \cos \psi \hat{j} - \frac{\partial}{\partial y} \cos \psi \hat{k} - \frac{\partial}{\partial z} \sin \psi \hat{i}$$

$$= \cancel{\frac{\partial}{\partial z} \hat{n}} + \left( \frac{\partial}{\partial x} \sin \psi - \frac{\partial}{\partial y} \cos \psi \right) \hat{k}$$

$$= \left( \cos \psi \frac{\partial \psi}{\partial x} + \sin \psi \frac{\partial \psi}{\partial y} \right) \hat{k}$$

$$= \hat{t} \cdot \left( \frac{\partial \psi}{\partial s} \hat{t} + \frac{\partial \psi}{\partial n} \hat{n} \right) \hat{k} = (\hat{t} \cdot \nabla_h \psi) \hat{k} = \frac{\partial \psi}{\partial s} \hat{k} = \frac{1}{R_s}$$

$$\nabla V \times \hat{t} = \begin{vmatrix} \hat{t} & \hat{n} & \hat{k} \\ \frac{\partial v}{\partial s} & \frac{\partial v}{\partial n} & 0 \\ 1 & 0 & 0 \end{vmatrix} = -\frac{\partial v}{\partial n} \hat{k}$$

$$\omega = V(\nabla \times \hat{t}) - \hat{t} \times \nabla V = V \frac{\partial \psi}{\partial s} \hat{k} + \frac{\partial v}{\partial n} \hat{k} = \left( \frac{V}{R} - \frac{\partial V}{\partial n} \right) \hat{k}$$

3. The approximate horizontal momentum equations considering Coriolis force, pressure gradient force and frictional force are represented as

$$\begin{aligned} \frac{Du}{Dt} &= \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = fv - \frac{1}{\rho} \frac{\partial P}{\partial x} + F_{rx} \\ \frac{Dv}{Dt} &= \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = -fu - \frac{1}{\rho} \frac{\partial P}{\partial y} + F_{ry} \end{aligned}$$

(a) Use above equations to derive the vorticity equation which could be expressed as:

$$\begin{aligned} \frac{d}{dt}(\zeta + f) &= \frac{\partial \zeta}{\partial t} + u \frac{\partial \zeta}{\partial x} + v \frac{\partial \zeta}{\partial y} + w \frac{\partial \zeta}{\partial z} + v \frac{df}{dy} = -(\zeta + f) \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) \\ &+ \frac{1}{\rho^2} \left( \frac{\partial \rho}{\partial x} \frac{\partial P}{\partial y} - \frac{\partial \rho}{\partial y} \frac{\partial P}{\partial x} \right) - \left( \frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) + \left( \frac{\partial F_r(y)}{\partial x} - \frac{\partial F_r(x)}{\partial y} \right) \end{aligned}$$

Ans:

$$\frac{\partial}{\partial y} \left( \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = fv - \frac{1}{\rho} \frac{\partial P}{\partial x} + F_{rx} \right) \quad (1)$$

$$\begin{aligned} (1) \Rightarrow & \left( \frac{\partial^2 u}{\partial y \partial t} \right) + \left( u \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial u}{\partial y} \frac{\partial u}{\partial x} \right) + \left( v \frac{\partial^2 u}{\partial y^2} + \frac{\partial v}{\partial y} \frac{\partial u}{\partial y} \right) + \left( w \frac{\partial^2 u}{\partial y \partial z} + \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) \\ &= - \left( \frac{1}{\rho} \frac{\partial p}{\partial x \partial y} + \frac{1}{\rho^2} \frac{\partial p}{\partial y} \frac{\partial \rho}{\partial x} \right) + \left( \frac{u \partial f}{\partial y} + f \frac{\partial v}{\partial y} \right) + \frac{\partial F_{rx}}{\partial y} \end{aligned}$$

$$\frac{\partial}{\partial x} \left( \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = -fu - \frac{1}{\rho} \frac{\partial P}{\partial y} + F_{ry} \right) \quad (2)$$

$$\begin{aligned} (2) \Rightarrow & \left( \frac{\partial^2 v}{\partial x \partial t} \right) + \left( u \frac{\partial^2 v}{\partial x^2} + \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} \right) + \left( v \frac{\partial^2 v}{\partial x \partial y} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} \right) + \left( w \frac{\partial^2 v}{\partial x \partial z} + \frac{\partial w}{\partial x} \frac{\partial v}{\partial z} \right) \\ &= - \left( \frac{1}{\rho} \frac{\partial p}{\partial x \partial y} + \frac{1}{\rho^2} \frac{\partial p}{\partial y} \frac{\partial \rho}{\partial x} \right) - \left( \cancel{u \frac{\partial f}{\partial x}} + f \frac{\partial u}{\partial x} \right) + \frac{\partial F_{ry}}{\partial x} \end{aligned}$$

$$\begin{aligned} (2) - (1) \Rightarrow & \boxed{\frac{\partial^2 v}{\partial x \partial t} - \frac{\partial^2 u}{\partial y \partial t}} + \boxed{u \frac{\partial^2 v}{\partial x^2} - u \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \frac{\partial u}{\partial x}} + \boxed{v \frac{\partial^2 v}{\partial x \partial y} - v \frac{\partial^2 u}{\partial y^2} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial v}{\partial y} \frac{\partial u}{\partial y}} \\ &+ \boxed{w \frac{\partial^2 v}{\partial x \partial z} - w \frac{\partial^2 u}{\partial y \partial z} + \frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z}} = \boxed{-\frac{1}{\rho^2} \frac{\partial p}{\partial x} \frac{\partial \rho}{\partial y} + \frac{1}{\rho^2} \frac{\partial \rho}{\partial x} \frac{\partial p}{\partial y}} - \boxed{v \frac{\partial f}{\partial y} + f \frac{\partial u}{\partial x} + f \frac{\partial v}{\partial y}} + \boxed{\frac{\partial F_{ry}}{\partial x} - \frac{\partial F_{rx}}{\partial y}} \end{aligned}$$

By the definition of vorticity  $\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$  and rearrange the equation, we have

R.H.S.

$$-(\zeta + f) \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \frac{1}{\rho^2} \left( \frac{\partial \rho}{\partial x} \frac{\partial P}{\partial y} - \frac{\partial \rho}{\partial y} \frac{\partial P}{\partial x} \right) - \left( \frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) + \left( \frac{\partial F_r(y)}{\partial x} - \frac{\partial F_r(x)}{\partial y} \right)$$

L.H.S.

$$\begin{aligned} & \boxed{\frac{\partial}{\partial t} \left( \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)} + \boxed{u \frac{\partial}{\partial x} \left( \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)} + \boxed{v \frac{\partial}{\partial y} \left( \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)} + \boxed{w \frac{\partial}{\partial z} \left( \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)} + v \frac{\partial f}{\partial y} \\ &= \boxed{\frac{\partial \zeta}{\partial t}} + \boxed{u \frac{\partial \zeta}{\partial x}} + \boxed{v \frac{\partial \zeta}{\partial y}} + \boxed{w \frac{\partial \zeta}{\partial z}} + v \frac{\partial f}{\partial y} \end{aligned}$$

From above, the vorticity equation can be expressed as

$$\begin{aligned} & \frac{\partial \zeta}{\partial t} + u \frac{\partial \zeta}{\partial x} + v \frac{\partial \zeta}{\partial y} + w \frac{\partial \zeta}{\partial z} + v \frac{df}{dy} \\ &= -(\zeta + f) \left( \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \frac{1}{\rho^2} \left( \frac{\partial \rho}{\partial x} \frac{\partial P}{\partial y} - \frac{\partial \rho}{\partial y} \frac{\partial P}{\partial x} \right) - \left( \frac{\partial w}{\partial x} \frac{\partial v}{\partial z} - \frac{\partial w}{\partial y} \frac{\partial u}{\partial z} \right) + \left( \frac{\partial F_r(y)}{\partial x} - \frac{\partial F_r(x)}{\partial y} \right) \end{aligned}$$

(b) In a similar fashion, please derive divergent equation:

$$\frac{\partial D}{\partial t} = \dots, \text{ where } D = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$$

Ans:

$$\frac{\partial}{\partial x} \left( \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} = f v - \frac{1}{\rho} \frac{\partial P}{\partial x} + F_{rx} \right) \quad (3)$$

$$\begin{aligned} (3) \implies & \left( \frac{\partial^2 u}{\partial x \partial t} \right) + \left( u \frac{\partial^2 u}{\partial x \partial x} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial x} \right) + \left( v \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial v}{\partial x} \frac{\partial u}{\partial y} \right) + \left( w \frac{\partial^2 u}{\partial x \partial z} + \frac{\partial w}{\partial x} \frac{\partial u}{\partial z} \right) \\ & = - \left( \frac{1}{\rho} \frac{\partial p}{\partial x \partial x} + \frac{1}{\rho^2} \frac{\partial p}{\partial x} \frac{\partial \rho}{\partial x} \right) + \left( u \cancel{\frac{\partial f}{\partial x}} + f \frac{\partial v}{\partial x} \right) + \frac{\partial F_{rx}}{\partial x} \end{aligned}$$

$$\frac{\partial}{\partial x} \left( \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} = -f u - \frac{1}{\rho} \frac{\partial P}{\partial y} + F_{ry} \right) \quad (4)$$

$$\begin{aligned} (4) \implies & \left( \frac{\partial^2 v}{\partial y \partial t} \right) + \left( u \frac{\partial^2 v}{\partial x \partial y} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right) + \left( v \frac{\partial^2 v}{\partial y \partial y} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial y} \right) + \left( w \frac{\partial^2 v}{\partial y \partial z} + \frac{\partial w}{\partial y} \frac{\partial v}{\partial z} \right) \\ & = - \left( \frac{1}{\rho} \frac{\partial p}{\partial y \partial y} + \frac{1}{\rho^2} \frac{\partial p}{\partial y} \frac{\partial \rho}{\partial y} \right) - \left( u \frac{\partial f}{\partial y} + f \frac{\partial u}{\partial y} \right) + \frac{\partial F_{ry}}{\partial y} \end{aligned}$$

By the definition of vorticity  $\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$  and rearrange the equation, we have

$$\begin{aligned} (2) - (1) \implies & \frac{\partial D}{\partial t} + D^2 + \frac{\partial u}{\partial x} D + \frac{\partial v}{\partial y} D + 2 \frac{\partial w}{\partial z} D \\ & = - \left( \frac{1}{\rho} \frac{\partial p}{\partial x \partial x} + \frac{1}{\rho^2} \frac{\partial p}{\partial x} \frac{\partial \rho}{\partial x} \right) + \left( f \frac{\partial v}{\partial x} \right) + \frac{\partial F_{rx}}{\partial x} - \left( \frac{1}{\rho} \frac{\partial p}{\partial y \partial y} + \frac{1}{\rho^2} \frac{\partial p}{\partial y} \frac{\partial \rho}{\partial y} \right) - \left( u \frac{\partial f}{\partial y} + f \frac{\partial u}{\partial y} \right) + \frac{\partial F_{ry}}{\partial y} \end{aligned}$$